CM3121 – Advanced Organic Chemistry

Dr. Zhu Ye

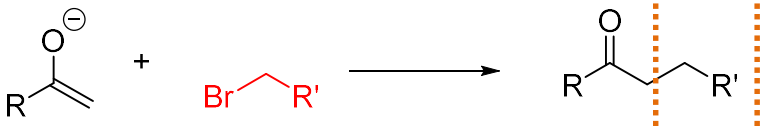
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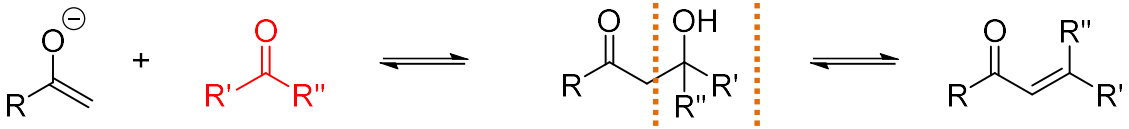
Enols and enamines

Reactions of enolates (basic conditions)

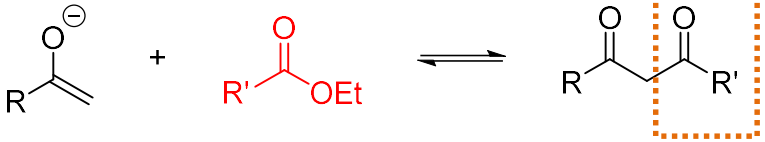
1. α-alkylation reaction



2. Aldol reactions: Enolate + aldehyde or ketone

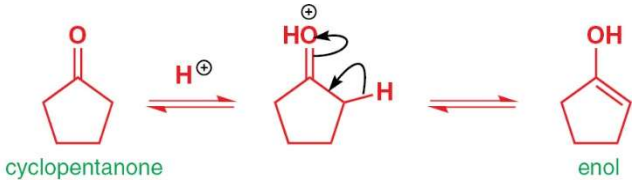


3. Claisen reaction: Enolate + ester

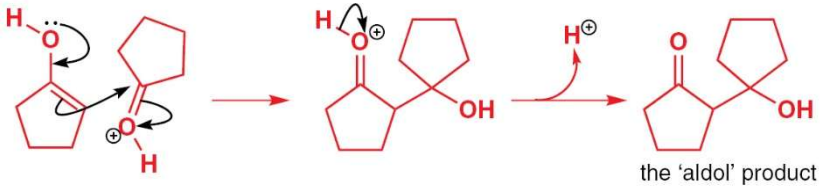


Mechanism of aldol reactions under acidic conditions

the acid-catalysed enolization step

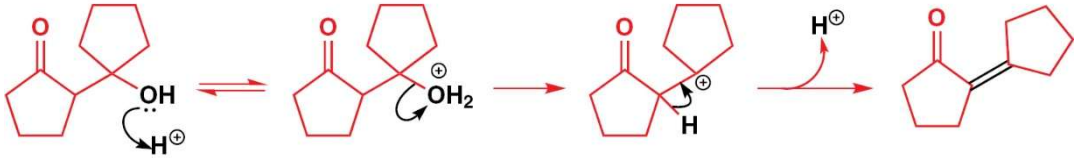


the acid-catalysed aldol addition step



Dehydration (difficult to avoid under acidic conditions)

the acid-catalysed dehydration step (E1 elimination)

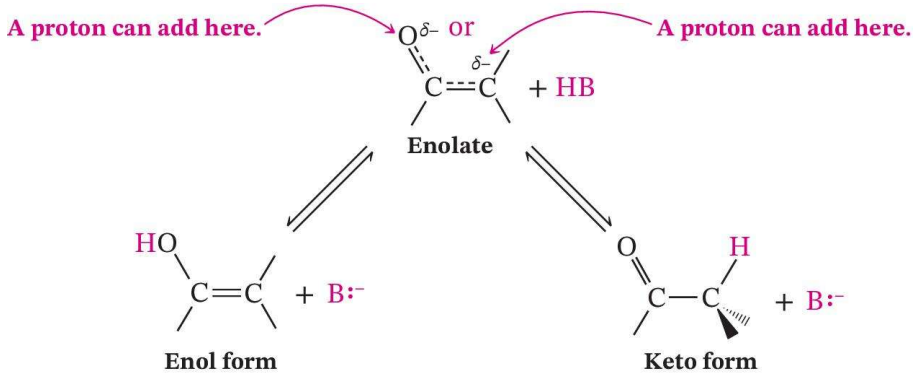


1. Keto and enol tautomers

- The keto and enol forms of carbonyl compounds are isomers, but of a special type: they are easily interconverted by H transfer in the presence of acid or base.

- Keto and enol forms are called tautomers, and their interconversion is called tautomerization.

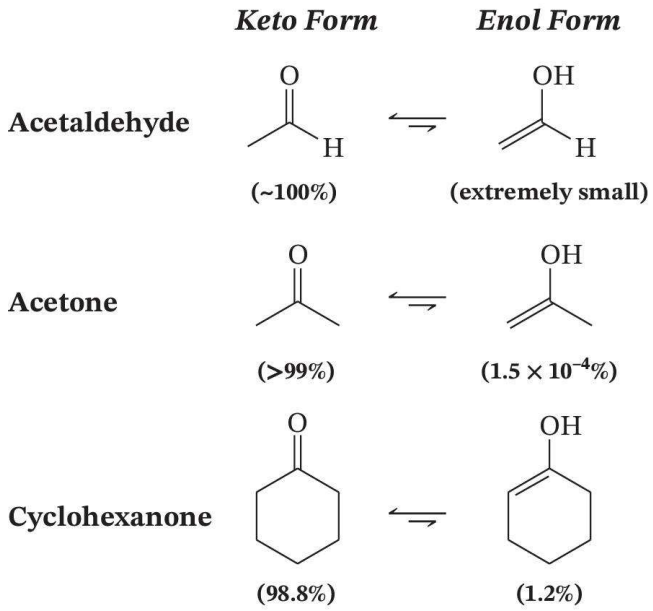
- When an enolate accepts a proton, it can accept a proton at the carbon to form the carbonyl compound (keto form) or at the oxygen to form enol (alkene alcohol).



Keto and enol tautomers

For simple mono carbonyl compounds, the amount of enol at equilibrium is very small

- C=O pi bond 364 kJ/mol
- C=C pi bond 250 kJ/mol



Single Bond Energies (kJ/mol of bonds)

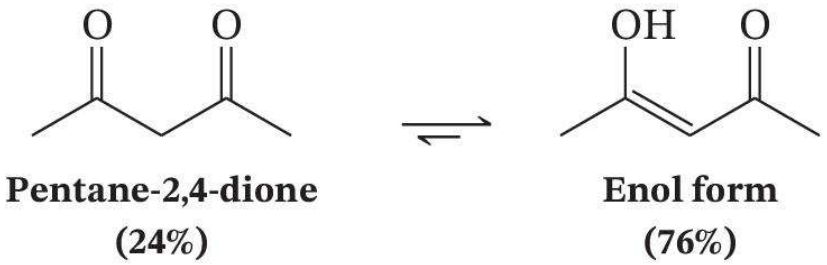
	H	C	N	O	S	F	Cl	Br	I
H	436								
C	413	346							
N	391	305	163						
O	463	358	201	146					
S	347	272	—	—	226				
F	565	485	283	190	284	155			
Cl	432	339	192	218	255	253	242		
Br	366	285	—	201	217	249	216	193	
I	299	213	—	201	—	278	208	175	151

Multiple Bond Energies (kJ/mol of bonds)

C=C 602	C=N 615	C=O 799
C≡C 835	C≡N 887	C≡O 1072
N=N 418	N=O 607	
N≡N 945	O=O 498	

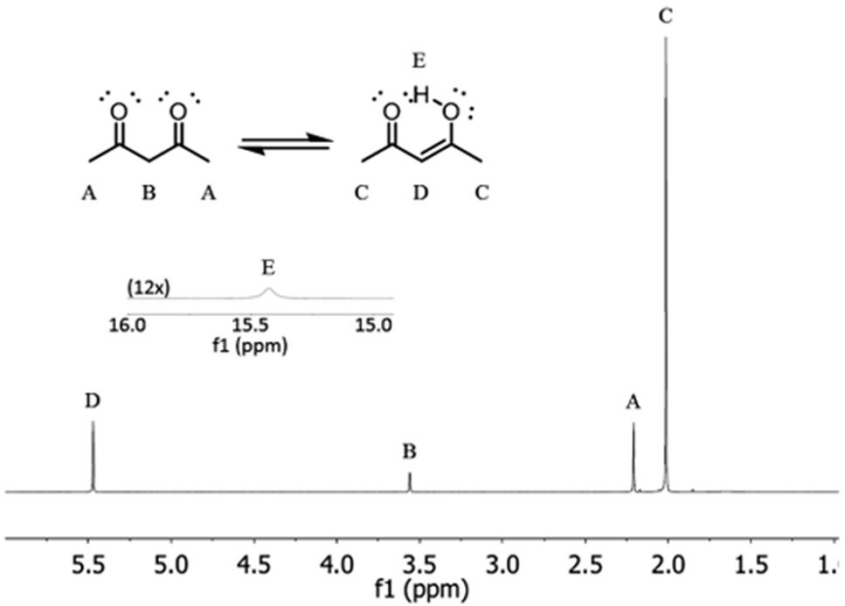
1,3-dicarbonyl compounds (stable enols)

The amount of enol present at equilibrium is far higher.

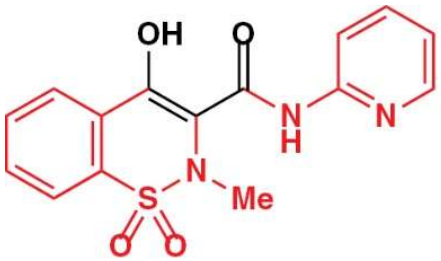


Pentane-2,4-dione
(24%)

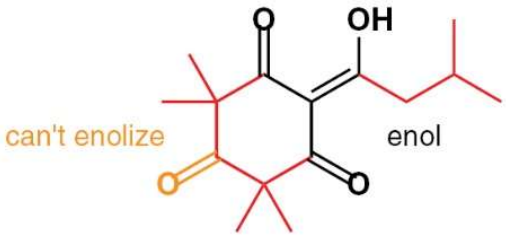
Enol form
(76%)



Examples of stable enols

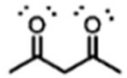
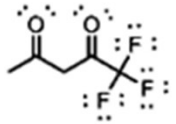
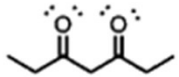
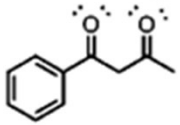
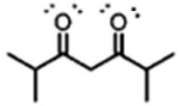
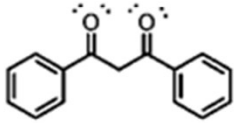
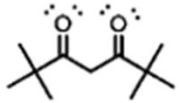
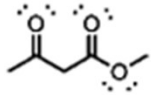
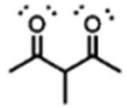
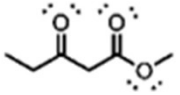
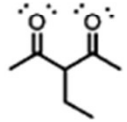
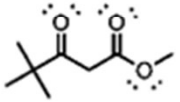
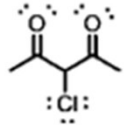
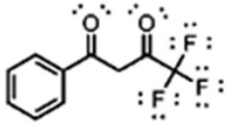
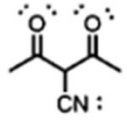
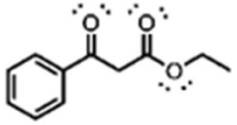


Piroxicam (Feldene®)
once-a-day treatment for arthritis



Leptospermone (Callisto®)
herbicide produced by the bottle-brush plant

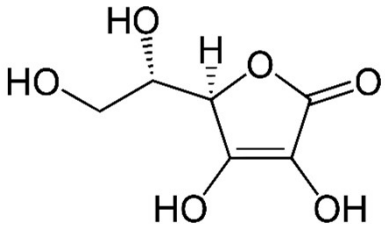
1,3-Dicarbonyl Compounds with Their Respective $K_{enol/keto}$ Values

#	Structure	$K_{e/k}$	#	Structure	$K_{e/k}$
1		5.5	9		$>10^4$
2		6.6	10		11.2
3		12.7	11		$>10^4$
4		38.0	12		0.095
5		0.83	13		0.047
6		0.29	14		0.19
7		22.3	15		$>10^4$
8		almost 100% enol[17]	16		0.26

Vitamin C

- Vitamin C is an essential nutrient involved in the repair of tissue, the formation of collagen, and the enzymatic production of certain neurotransmitters. It is required for the functioning of several enzymes and is important for immune system function. It also functions as an antioxidant.

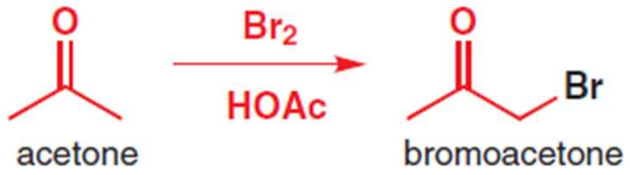
- Most animals are able to synthesize their own vitamin C. However, humans must acquire it from dietary sources.



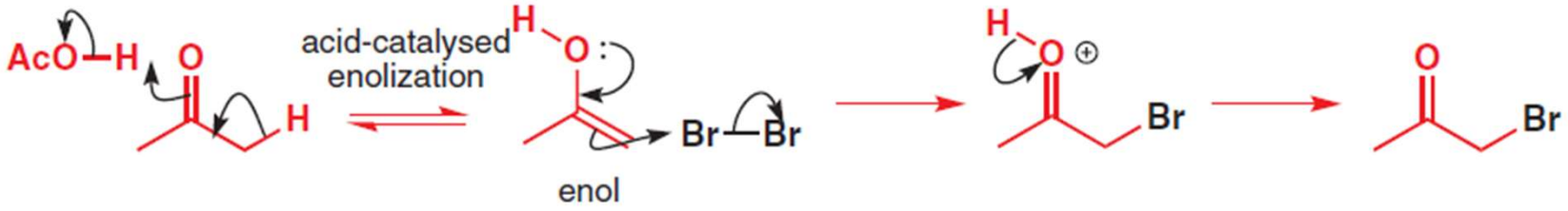
- Vitamin C (also known as ascorbic acid and ascorbate) is a water-soluble vitamin found in citrus and other fruits and vegetables.

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2. Halogenation reactions (under acidic conditions)

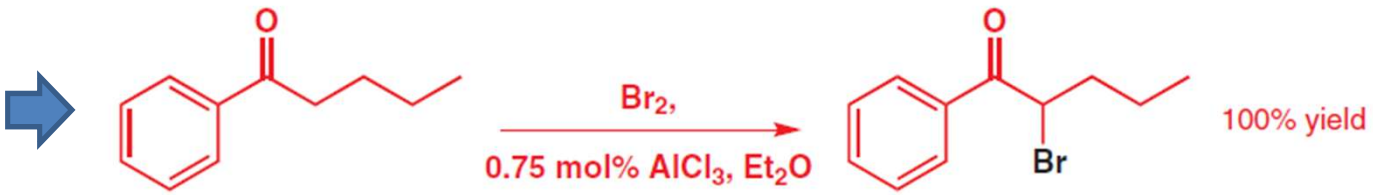


Mechanism (acidic conditions):

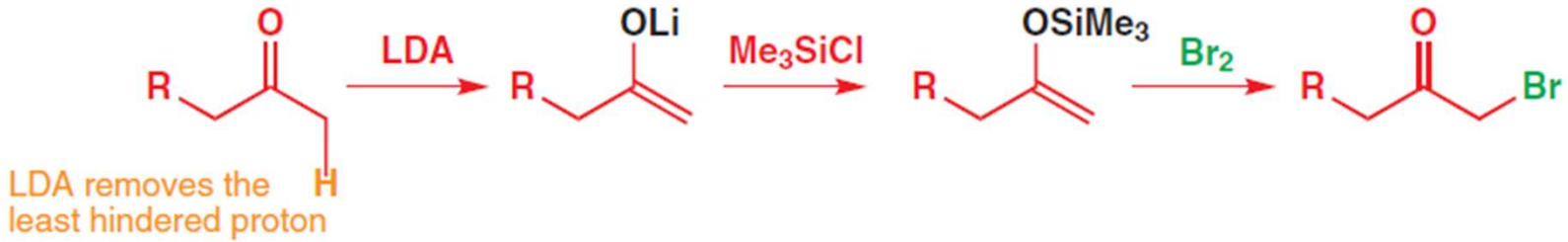


Reactions in the presence of Lewis acid:

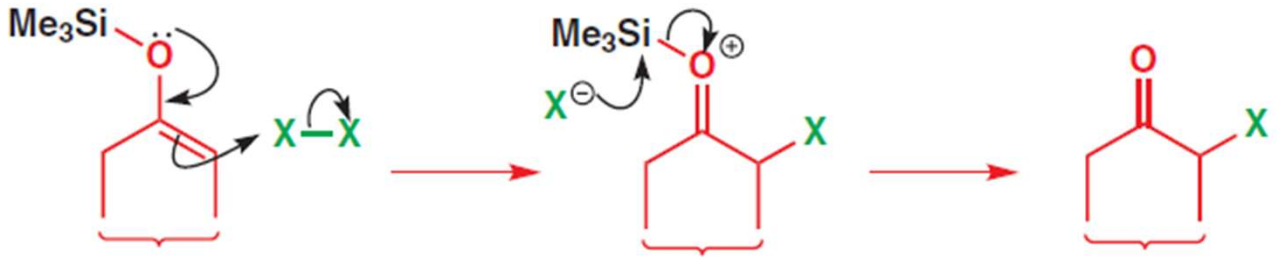
Deactivated



Halogenation reactions (through silyl enol ethers)

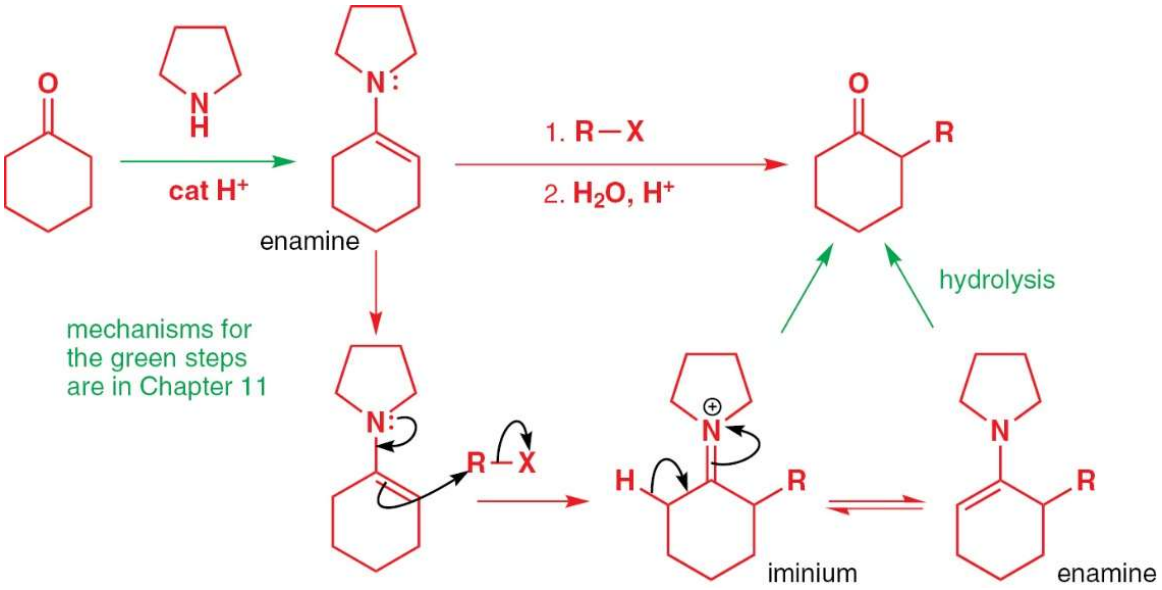


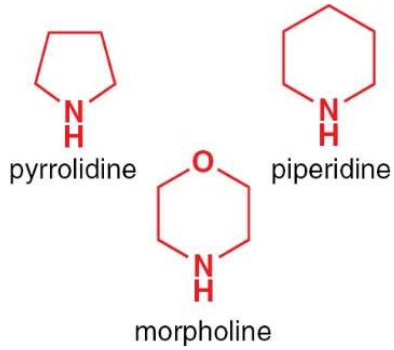
Mechanism (silyl acts as a “large hydrogen”):



3. Enamines

Enamines are formed when aldehydes or ketones react with secondary amines.





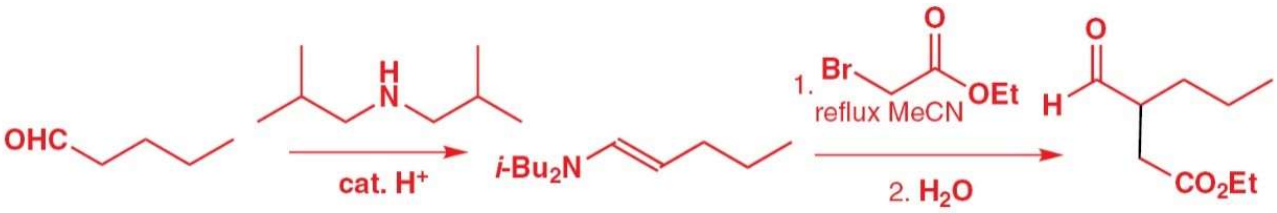
secondary amines

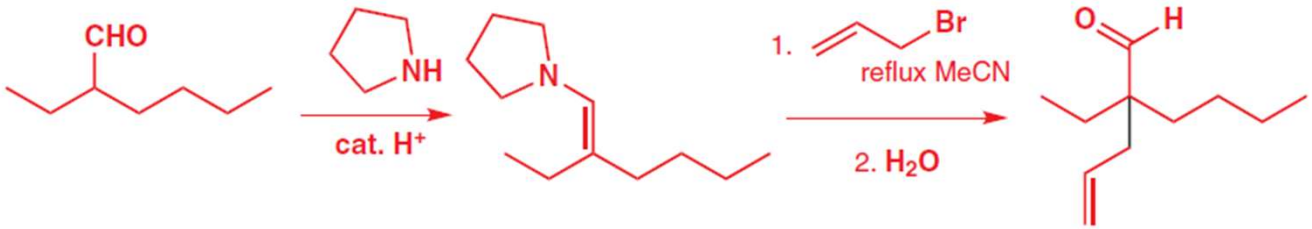
Imine and Enamine are tautomers

Recall: Reductive amination

Enamines in alkylation

Enamine provide good solution for alkylation of **aldehydes**: α-alkylation can occur without strong base (no self-condensation)





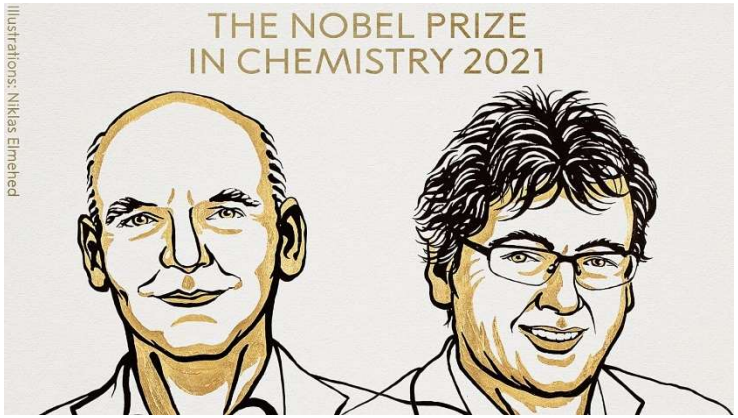
Enamines in aldol reactions

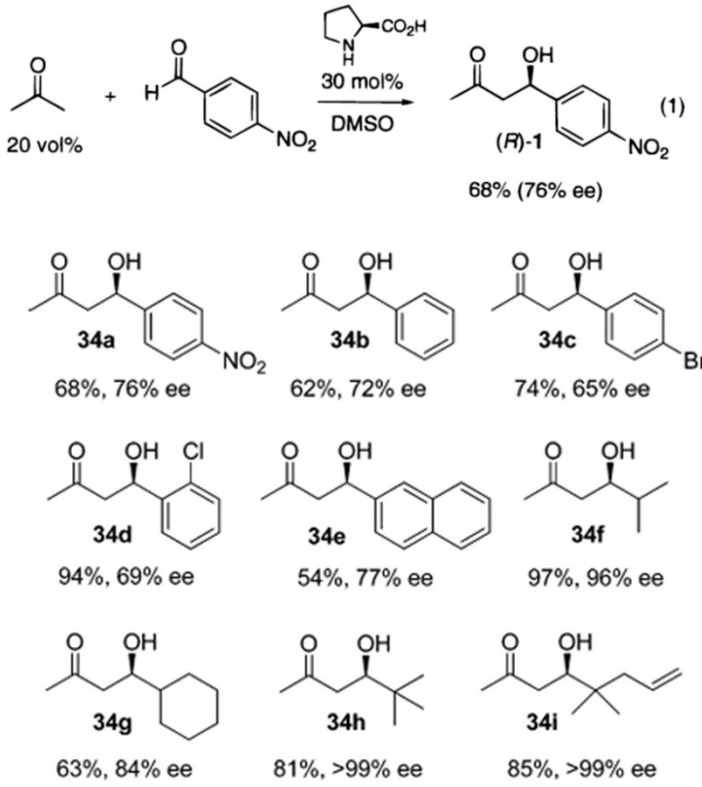
Proline-Catalyzed Direct Asymmetric Aldol Reactions

Benjamin List,* Richard A. Lerner, and Carlos F. Barbas III

The Skaggs Institute for Chemical Biology and the Department of Molecular Biology The Scripps Research Institute 10550 North Torrey Pines Road, La Jolla, California 92037

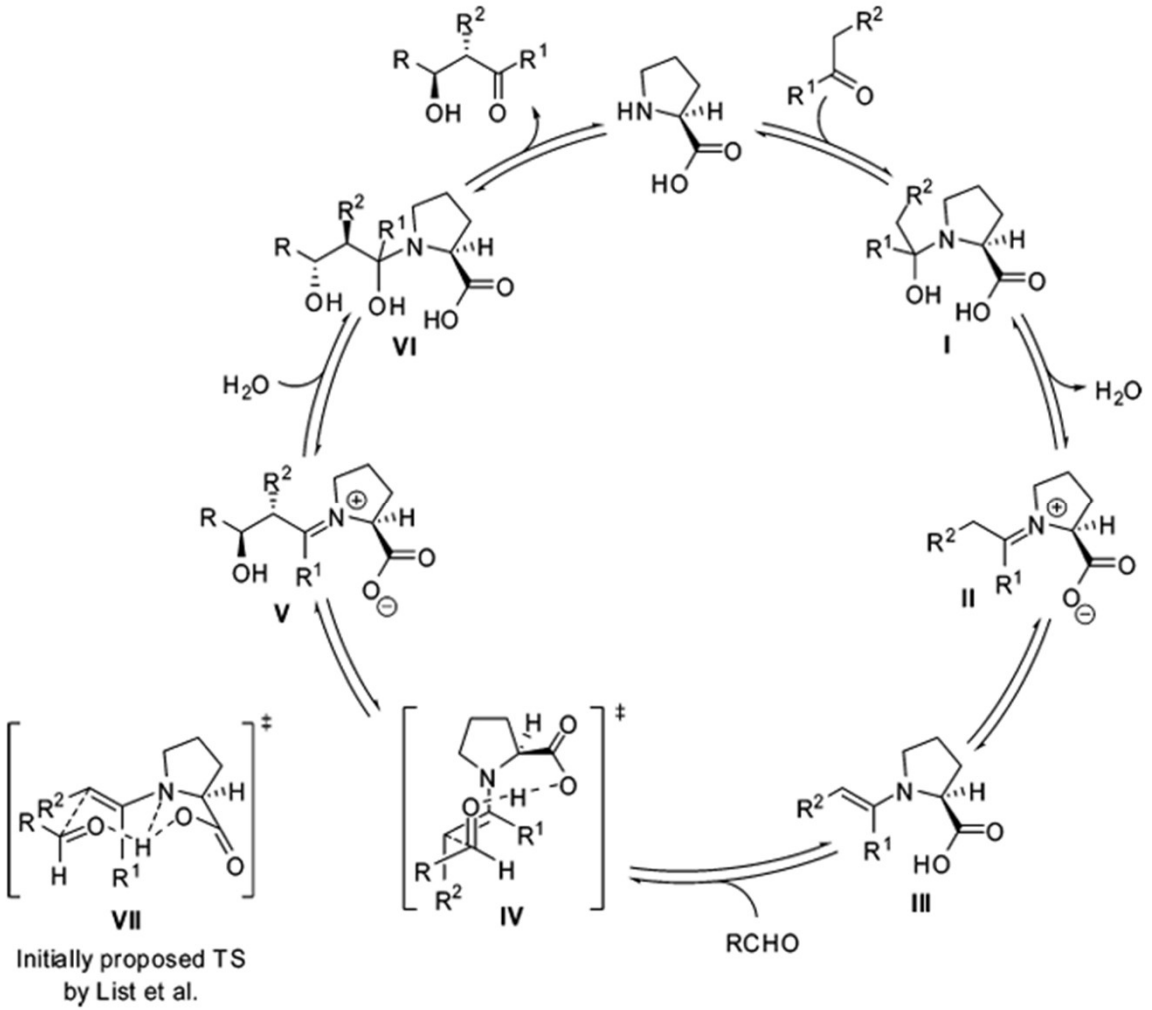
Benjamin List and David MacMillan are awarded the Nobel Prize in Chemistry 2021 because in 2000 they, independent of each other, developed a third type of catalysis. It is called asymmetric organocatalysis and builds upon small organic molecules.





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Enamines in asymmetric aldol reactions



Summary

- Keto and enol are tautomers
- 1,3-dicarbonyl compounds form stable enols
- Halogenation reactions of enols
- Enamines as nucleophiles for alkylation and aldol-type reactions